Smit Bhavsar

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SUMMARY OF QUALIFICATIONS

- AI SOC Design Verification Engineer with **3+ years** of experience in validating complex components, specializing in hardware accelerators, **formal verification**, SOC **Coverage**, Chiplet Co-Simulation & **Register Metadata** and **PPR** (Processor Programming Reference).
- Expertise includes leveraging **UVM-based testbenches**, **automating** verification workflows, and improving **design validation** processes. This involves applying Formal Verification wherever UVM (Universal Verification Methodology) is present.
- Proficient in scripting languages such as **Python, Perl, SystemVerilog** and **Ruby**, utilizing these skills to enhance test environments, automate **regression testing**, and streamline verification, resulting in up to a **60% reduction** in test failures.
- Strong **cross-functional collaborator**, working with teams in India, USA, and China, contributing to the successful development and release of high-impact products like the **AMD Instinct™ MI300 Series Accelerators** and the latest AMD MI Instinct: **MI400/MI450 Series AI Accelerators**. MI450 was announced to be used in **AMD x OpenAI** partnership.

EDUCATION

Toronto Metropolitan University, Toronto, Ontario (Ryerson University)
Bachelor of Engineering (B.Eng) in Computer Engineering, 2019-2024
Cumulative CGPA: **3.52**, Dean's Honour List 2019-2024

WORK EXPERIENCE

AMD, AI & ML SOC Design Verification Engineer 2 | May 2024 – Present

- **Collaborated** with **cross-functional teams** in India, Boston and China to **design** and **implement testcases** for validating the connectivity of various **SOCs** and catching several **RTL** related **bugs** and issues, and using **UVM** (Universal Verification Methodology) based testbenches for digital designs, **increasing test coverage by 35%** and **improving** overall **verification efficiency by 20%** compared to previous methodologies (Manual / Direct testing).
- Wrote and executed **over 500 test cases** for functional, performance, memory and power verification, achieving **99% test coverage** and identifying critical design flaws, which included but not limited to various connectivity checks and functional tests, along with the implementation of **Formal Verification** in our SOC environment.
- **Collaborated** closely with **RTL** engineers to **implement advanced** logic designs, resulting in a **25% increase in clock gating efficiency** and a corresponding improvement in power and memory consumption metrics using **LSF** to monitor.
- Led the adoption of advanced formal verification methodologies, **reducing** post-silicon **debugging time by 50%.**
- Applied knowledge of **Formal Verification** protocols such as Formal **CC, FPV and SEQ** on **AXI** and **PCIe** Subsystems to validate SOC functionality, ensuring compliance with industry standards and specifications.
- Successfully identified and **debugged** design issues by performing functional and formal verification, contributing to a **15% reduction in pre-production bugs**.
- Utilized scripting languages including **Python, Perl**, and **Ruby** to **automate** verification tasks, streamlining workflow processes and **reducing manual effort by 25%.**
- Utilized **Synopsys Verdi** for advanced debugging, **analyzing waveforms**, **log files**, and microcode **trace dumps** to identify root causes of design failures.

AMD, SOC Design Verification & Infrastructure Engineer Intern | May 2022 – August 2023

- Improved, modified and maintained a regression system with **Perl, Python/Ruby** and **MySQL** used by teams across different time zones to add, update, dispatch and review tests on the latest design and frameworks.
- Advanced scripting using Perl and Ruby/Python to **automate** several aspects of the test bench, including maintaining a **sanity/regression dashboard "smart" system**, and validating incoming **Perforce** changes.
- Developed **test cases** for verifying the functionality of exclusive features by partnering with **SOC and IP teams**.
- **Reduced** amount of sanity/regression test failures **by 60% per week** (down from **10** a week to **4** a week on average) by implementing various debugging techniques, and launching a new failure_analysis.rb script to automatically filter failures.
- Used Python/Ruby and bash scripting, developed an automation script that **creates a sim build** on a **cloud server**, **transfers build** to the specified system, **performs regression testing**, renames and saves all the log files.
- Created a **disk monitor script** in Python/Ruby, displaying **disk usage** of all our regression disks in a certain moment in time, made entirely on our Linux environment.
- Used **Confluence and JIRA** for ticket and task tracking management.

PROJECT EXPERIENCES

AMD Instinct™ EPYC MI430X Series Accelerators | October 2025 – Present

- Recently announced to be used apart of the new Meta AI platform.
- Acted as a technical multiplier by transferring knowledge from earlier programs, helping teams avoid redundant verification effort and **accelerating overall schedule predictability** for MI430X milestones.
- Solely owned **register metadata review** and debug for MI430X, triaging inquiries from multiple IP teams by analyzing **byte-offset and address-mapping files** to resolve mismatched register addresses; authored the **register specification** document and evolved it into a GUI-driven, and automated generation flow.
- Owned **end-to-end SoC functional, toggle, and tile coverage** for **1 of 3 MI430X chipsets**, designing and deploying a new **object-oriented coverage flow with a scalable waiver system**, which reduced manual waiver effort by **~40%** and enabled faster iteration and extensibility of coverage models.
- Led **cross-functional coverage alignment** by hosting and driving recurring meetings with **Power, MMHUB, SMU, Emulation, and verification teams**, accelerating coverage closure and achieving **98% total SoC coverage within 3 months** of ownership handoff.
- Assisted in writing and reviewing multiple **MI430X Processor Programming Reference (PPR) documents**, while concurrently closing **10+ Jira tickets** across verification, coverage, and infra tasks during a high-pressure development phase, demonstrating strong execution, prioritization, and cross-team collaboration.

AMD Instinct™ EPYC MI400/MI450 Series Accelerators | May 2024 – October 2025

- Worked on the next generation **Series Accelerator** whilst working full-time at AMD, used a part of AMD x OpenAI partnership in late 2026.
- **Developed** and maintained modular, **reusable verification components**, allowing for efficient testbench configuration and **reducing setup** time for new verification environments **by 50%.**
- Spearheaded the integration of **formal verification** methods with **UVM** simulation-based environments, resulting in a **40% reduction** in time spent on simulation for verifying **complex design** constraints and properties.
- Implemented **UVM** and **formal verification** techniques, increasing testbench **efficiency by 35**% by automating previously manual and direct testing processes in areas such as power, memory consumption, SMU, and other functional/IP aspects of our SoC. This adoption led to significant improvements across all areas of the SoC.
- **Applied clock gating** to **reduce** dynamic power consumption in specific SoC components by selectively **disabling** clock signals for **inactive sections**. Utilized **logic gates** to control clock paths, contributing to optimized power savings and energy efficiency within the system.
- Introduced formal verification techniques (primarily **SEQ**-based verification) to optimize the verification of two similar PCIe subsystems. One subsystem was fully verified through traditional manual and **UVM** design methods, while the other had not yet been verified. By leveraging common data paths.
    - **Reduced verification time by 80%** for similar PCIe subsystems through the use of formal verification, speeding up the overall verification process.
    - **Trained** and mentored **RTL engineers** on formal verification methodologies, enabling them to incorporate formal tools into their workflows for more efficient design verification.
    - Presented findings at **AMD case competitions** and **innovation showcases**

AMD Instinct™ EPYC MI300 Series Accelerators | May 2022 – August 2023

- Worked on the **MI300 Series Accelerator** whilst interning at AMD.
- Collaborated with cross-functional teams (**Infra, SMU, PMU, DFT, FPGA**, etc.) to address **pre-silicon issues** and integrate the **MI300** chip into the target platform seamlessly.
- Actively participated in **5 cross-functional team projects**, collaborating with colleagues from different departments to achieve common objectives for the overall design.
- **Automated 10 manual test procedures** using **Python** scripting, resulting in a **30% reduction** in testing time and increased efficiency in the verification process.
- Assisted the **design verification** and **infrastructure team** in executing test plans for the MI300 chip, contributing to a **10% increase in verification coverage**.
- Identified and **reported 25+ bugs** during **regression testing**, leading to their **timely resolution** and **improvement** in the overall stability of the **MI300 design**.
- Supported infra maintenance activities, including **updating documentation** and organizing **test benches**.
- Conducted **performance benchmarking** on the MI300 chip, collecting and **analyzing data** from **50+ tests** to provide insights into its computational capabilities.